Auditing PPI Graph Augmentation for Missense Variant Pathogenicity Prediction under Topology, Connectivity and Homology Constraints

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Abstract—We audit fixed two-hop protein-feature propagation (FPP) for missense variant pathogenicity prediction using 177,213 ClinVar variants, an external UniProt variant source and two networks. FPP has no learned graph parameters. Matched STRING controls show sensitivity to connectivity and protein–topology correspondence. STRING fusion improves over raw-embedding fusion on the original split, whereas its homology-aware gains over strong variant scores remain uncertain. On 10,785 external variants, STRING increases mean AUC from 0.6932 to 0.7304; the ensemble gain is +0.0375 with a protein-bootstrap 95% interval of [+0.0173, +0.0564]. A stricter nested subset supports the same direction. BioGRID does not establish external gains over raw embeddings or incremental fusion gains over raw-embedding fusion. These results support external-source transfer for STRING under the stated exclusions and sensitivity to network choice. They concern transductive protein-context features, not within-protein substitution discrimination or the performance of trained graph models.

Index Terms—Missense variant pathogenicity, STRING networks, fixed feature propagation, protein language models, evaluation audit.

I. Introduction

Missense variant pathogenicity prediction has been advanced by established methods such as CADD, REVEL, MVP, PrimateAI, EVE and AlphaMissense [5], [7], [9], [12], [18], [23], as well as by protein language models such as ESM [14]. These approaches capture sequence, evolutionary and annotation-derived signals. Disease-associated proteins are also embedded in molecular interaction networks, and disease-associated mutations can perturb protein-protein and other macromolecular interactions [1], [3], [16], [19]. PPI graphs provide an additional source of context for missense variant prediction.

Earlier approaches incorporated protein-interaction and other molecular-network features into single-amino-acid variant prioritization, showing that network context can provide information complementary to sequence, structure and annotation features [17], [26]. More recent graph-based variant predictors include graph attention models for missense effects, heterogeneous gene-variant graph approaches, multimodal variant graph methods and disease-specific graph models [4], [8], [27], [28]. These studies establish the relevance of relational structure to variant interpretation. We focus on how to interpret gains obtained by propagating protein language model embeddings over a shared STRING association graph before variant-level classification.

A shared-graph setting mixes several effects. Train, validation and test variants may have distinct protein identifiers while their proteins remain directly connected in the same STRING graph. Related proteins may also cross partitions when homology is not explicitly controlled [15], [25]. Strong variant-effect predictors such as CADD and AlphaMissense can further reduce the marginal contribution of graph propagation. We evaluate protein-topology correspondence, graph connectivity, protein-level uncertainty, homology-aware performance and complementarity to established predictors.

We audit raw ESM, fixed propagation, external scores and fusion models on a reconstructed ClinVar-STRING dataset. The contribution is an evaluation study combining graph-intervention controls, matched fusion, protein-clustered uncertainty, external-source transfer and alternative-network evaluation. Topology and error-transition profiles are exploratory. The audit does not introduce a new predictor or cover trained message-passing architectures.

II. Materials and Methods

A. Dataset and graph construction

ClinVar variant annotations were obtained from the official NCBI ClinVar FTP tab-delimited variant-summary file [13]. All experiments used archived, checksum-fixed processed tables; the exact historical source-release date could not be independently verified. We reconstructed a filtered, non-conflicting missense dataset and mapped variants to STRING human protein identifiers. We retained GRCh38 germline missense SNVs with criteria-supported review status, combined pathogenic and likely pathogenic records into the positive class, combined benign and likely benign records into the negative class, removed uncertain or conflicting records, and collapsed duplicate VariationIDs. The final ClinVar-STRING dataset contained 177,213

missense variants mapped to 14,123 labelled STRING proteins. The propagation graph contained 19,622 STRING protein nodes and 13,715,404 directed edge-index entries from the STRING v12 protein-association network [24]. We used this network as STRING network context; its association evidence extends beyond direct physical interactions.

B. Protein representations and fixed propagation

Raw protein features were generated using the ESM-2 650M checkpoint, with 1,280-dimensional protein-level vectors obtained by residue averaging [14]. Fixed protein-feature propagation (FPP) was computed by two-hop GCN-style propagation over the normalized STRING graph [11]. FPP has no learned graph-convolution parameters; the downstream MLP alone was trained. Fixed propagation isolates topology-induced smoothing from capacity gains due to trainable graph layers and provides a controlled representation for the audit. Let X denote the raw ESM protein-representation matrix. The matched implementation adds self-loops before symmetrizing the retained edge list and sums duplicate contributions when coalescing. Thus $\tilde{A} = (E + I) + (E + I)^{\mathrm{T}}$, where E is the edge-index count matrix; STRING combined scores are not propagation weights. Let $\tilde{D}$ denote its weighted degree matrix. We computed the symmetrically normalized adjacency matrix as

$$\hat{A} = \tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}, \tag{1}$$

$$H = \frac{1}{3} \left(X + \hat{A}X + \hat{A}^2 X \right). \tag{2}$$

The fixed two-hop FPP representation H averages the original ESM features, one-hop and two-hop STRING-smoothed features. Propagation used only graph structure and ESM features; downstream classification used the representation of each variant's mapped STRING protein. The full graph contained train, validation and test nodes with their ESM features, making the evaluation transductive. Held-out features and graph structure were available during propagation, but held-out labels were not used. Embedding-only models assign identical inputs to variants mapped to the same protein; they cannot resolve residue-specific effects within that protein.

C. Splits, classifiers and external predictors

The original split assigned train, validation and test variants by STRING protein ID, with no shared proteins across partitions. The split contained 141,864 training variants, 17,129 validation variants and 18,220 test variants. The homology-aware split used MMseqs-derived protein clusters, assigned each cluster wholly to one partition by greedy balancing of variant-row counts, without label-prevalence stratification, and preserved the same row counts [22]. We use this split as a stress test. Its test positive rate was 10.56%, compared with 31.95% in the original split, which limits direct cross-split AUPRC comparison.

All current comparisons used downstream MLPs with 512/128 hidden units, dropout 0.3, Adam optimization and validation-AUC checkpoint selection [10]. Classifier seeds 42–44, a maximum of 50 epochs and patience five were shared across arms; permutation controls crossed three graph permutations with the same classifier seeds. CADD v1.7 and AlphaMissense scores were matched to variant records [5], [21]. Missing values were imputed using training-set medians, with an AlphaMissense missingness indicator; inputs were not standardized. Models shared hidden layers and preprocessing, although score-only models had fewer input weights. Training hyperparameters, score coverage and historical protocols are detailed in Supplementary Methods 3 and 5 and Tables 1 and 8.

D. External-source and second-network validation

Frozen original-split raw-ESM, STRING-FPP and BioGRID-FPP classifiers were evaluated on UniProtKB/Swiss-Prot HUMSAVAR 2026_03 without external fitting or tuning [29]. Canonical sequences had to match a unique frozen graph node exactly, with the reference residue verified. We excluded original variants using the union of accession-, node- and gene-plus-substitution keys and dbSNP identifiers. The primary set excluded original training/validation proteins and identical full-length sequences: 10,785 variants from 2,498 proteins. A nested sensitivity set also excluded original test proteins and identical sequences: 6,934 variants from 1,713 proteins. All remained nodes of the frozen transductive graph; these exclusions do not establish patient, evidence or full homology independence.

BioGRID 5.0.262 human physical-category records [30] were filtered to remove RNA-related methods and ambiguous mappings, then mapped to the same 19,622 nodes. The graph contained 931,458 unique undirected edges and 1,402 isolates; all test rows were retained. Nine additional classifiers covered BioGRID representation-only prediction and original/homology fusion, using the same operator, seeds and training budget. Of its edges, 56.54% also occurred in STRING. Paired protein-bootstrap intervals used 2,000 resamples of three-seed mean probabilities, conditional on fitted models and without multiplicity adjustment. Full filtering, run verification and AUPRC results are in Supplementary Methods 6 and Tables 14–17.

E. Audit analyses

Uncertainty was assessed using variant-level bootstrap and protein-clustered bootstrap [6]. Connectivity tests removed direct train-test STRING edges and regenerated FPP embeddings. Partial edge-ablation curves removed 0%, 25%, 50%, 75% and 100% of direct train-test protein pairs using random removal, higher-STRING-score-first removal and lower-STRING-score-first removal. Topology

TABLE I

Run provenance for the main results. Historical, matched and external-validation protocols are in Supplementary Tables 8, 11 and Methods 6. Runs count fitted classifiers, not bootstrap resamples.

Evidence	Runs	Summary used
Fig. 1B,D/F representations	3/arm	Matched seeds/operator/budget
Fig. 1C uncertainty	Ensemble	Paired variant/protein bootstrap
Fig. 1E partial ablation	Historical	Unequal protocols; descriptive
Fig. 2B fusion controls	3/arm	Matched seeds/operator/budget
Fig. 2C fusion intervals	Ensemble	Protein-clustered bootstrap
Fig. 2E homology fusion	3/arm	Matched score/raw/FPP arms
Fig. 2D; Suppl. Figs. S1–S2	1/arm	Historical retained predictions
Fig. 3A external source	3/arm	Frozen checkpoints; no tuning
Fig. 3B BioGRID fusion	3/arm	Each split; six fusion fits

Each permutation arm uses 3 graphs by 3 classifier seeds (9 fits). The 48 matched audit fits and 9 additional BioGRID fits (3 representation, 6 fusion) allow at most 50 epochs. Bootstrap intervals condition on fitted predictions; permutations are not 9 independent graph samples.

permutation preserved the unlabeled graph structure and randomized the correspondence between protein identity and graph position, testing whether generic graph smoothing explained the gain. Fusion controls concatenated CADD+AlphaMissense features with raw ESM, edge-ablated FPP and topology-permuted FPP representations, testing representation complementarity and sensitivity to protein-topology correspondence. Edge removal also changes normalization and retains indirect cross-partition paths; it does not isolate a causal contribution of direct connectivity.

The implementation audit identified a self-loop convention mismatch between historical full FPP and its graph controls. Current graph controls were regenerated with the full-FPP operator; the saved full-FPP representation was numerically checked. Historical controls remain separately labeled; environment restoration and numerical verification are documented in Supplementary Methods 5.

Exploratory analyses stratified retained-reference predictions by degree, k-core, betweenness, clustering and neighbour-risk features, and examined corrected and degraded predictions. Neighbour-risk features used training labels only. Feature-wise comparisons used false-discovery-rate correction [2], but selected subgroup intervals did not account for test-set selection. AUPRC was interpreted within each split because precision-recall metrics depend on class prevalence [20].

III. Results

A. Full-graph FPP gains and connectivity sensitivity

Table I separates new matched fits from historical retained predictions. With classifier seeds 42–44, raw ESM reached test AUC 0.7835 ± 0.0063 and AUPRC 0.6116 ± 0.0066; full FPP reached AUC 0.7950 ± 0.0016 and AUPRC 0.6484 ± 0.0118 (Fig. 1B). On ensemble-averaged predictions, the FPP-minus-ESM AUC difference was +0.0107, with protein-clustered 95% CI [-0.0218, +0.0426]; the AUPRC difference was +0.0413, CI [-0.0109, +0.0770] (Fig. 1C). Variant resampling ignores within-protein dependence and gives a different uncertainty estimate. These intervals condition on the three fitted classifiers, rather than incorporating retraining variability.

Removing 1,242,422 direct train-test edge-index entries (9.06%; 621,211 unique undirected pairs) and recomputing the same normalized operator gave edge-ablated FPP AUC 0.7702 ± 0.0077 and AUPRC 0.6075 ± 0.0139 (Fig. 1D). Repeating with three topology permutations by the same three classifier seeds gave pooled AUC 0.7803 ± 0.0092 and AUPRC 0.6102 ± 0.0085 (Fig. 1F). Per-graph and paired-seed results are retained separately in Supplementary Table 11. Matched protocols remove the previous seed, budget and self-loop mismatches, but edge removal still changes degrees and indirect connectivity remains; the contrast does not identify a purely causal contribution of direct paths.

The historical partial-ablation curve (Fig. 1E) used different classifier seeds and budgets, with reused endpoints and the old control operator. It remains an exploratory dose-response description, not part of the new matched effect estimate.

B. Architecture-matched fusion and homology-aware stress test

On the original split, mean AUC was 0.9572 for CADD+AlphaMissense, 0.9682 with raw embeddings and 0.9716 with FPP (Fig. 2B). The ensemble FPP-minus-raw fusion margin had a protein-clustered 95% interval of [+0.0008, +0.0061]. The edge-ablated margin remained uncertain, whereas the permutation margin was small and positive conditional on the sampled graphs (Fig. 2C; Supplementary Table 12). These exploratory intervals are not adjusted for multiple comparisons and condition on fitted models; an interval spanning zero does not establish equivalence.

Figure 2D retains historical homology-aware representation estimates. The current homology-aware fusion comparison gave mean AUC 0.9755 for CADD+ AlphaMissense, 0.9727 with raw embeddings and 0.9752 with FPP, using three matched seeds per arm (Fig. 2E). The ensemble FPP-minus-score AUC difference was +0.0001, with protein-clustered 95% CI [-0.0019, +0.0021]; the AUPRC interval also spanned zero (Supplementary Table 12). Thus, repeated fitting establishes neither benefit nor consistent harm from STRING FPP relative to scores alone. Mean run metrics and metrics of averaged probabilities can give different close rankings. Redundant high-dimensional inputs, additional input weights, lack of feature scaling and altered protein composition could

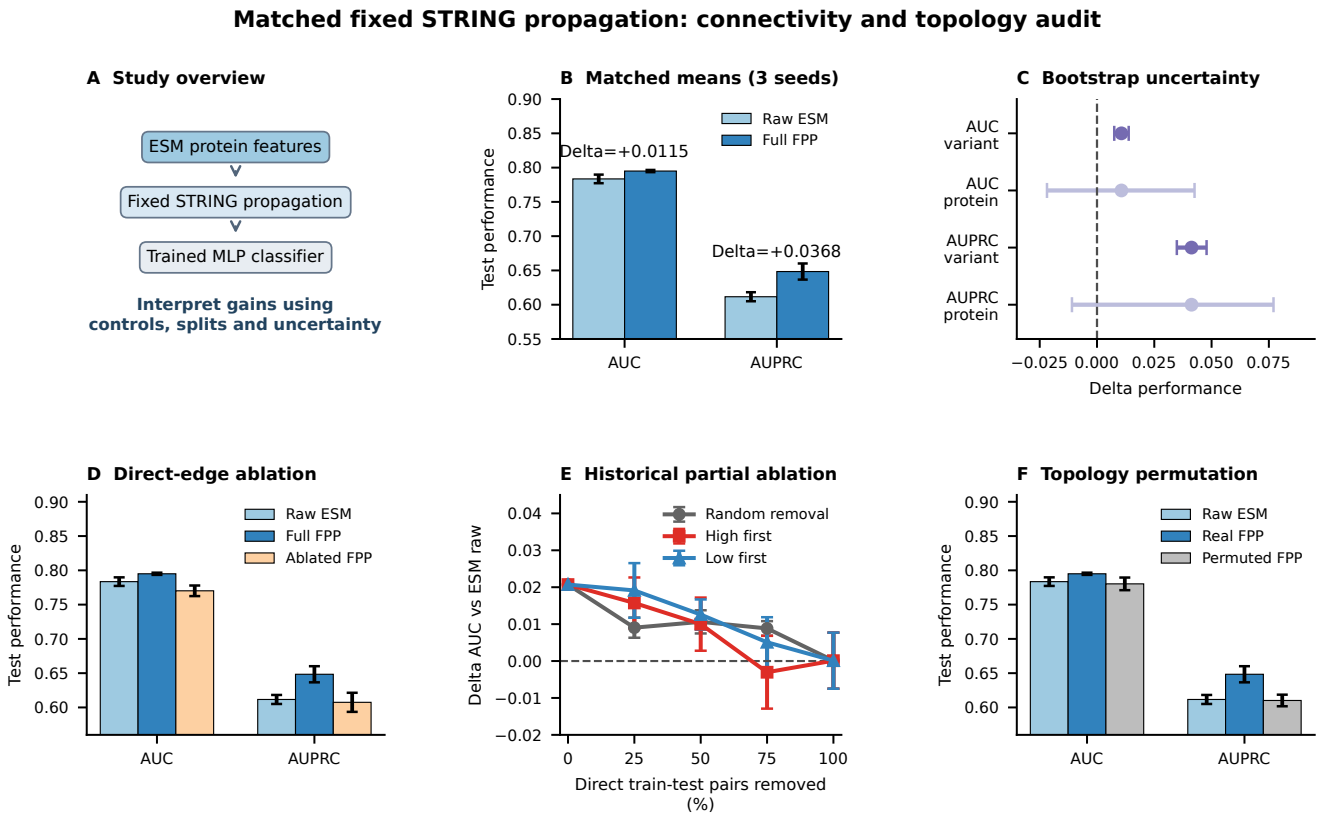


Fig. 1. Matched audit of fixed STRING propagation. A, Study overview. B, Raw ESM and full FPP: three classifier seeds per arm, mean and SD. C, Paired bootstrap of ensemble-averaged predictions, comparing variant and protein resampling; 2,000 replicates each. D, Direct-edge ablation with the common propagation operator and matched classifier protocol, mean and SD across three seeds. E, Historical partial-ablation curve with unmatched protocols and reused endpoints; it is not a matched continuation of panel D. F, Three topology permutations by the same three classifier seeds; pooled mean and SD describe both sources of dispersion. New arms share maximum 50 epochs and validation-based early stopping.

contribute to this behavior; these explanations were not experimentally separated. AUPRC comparisons remain within-split because prevalence differs.

C. External-source transfer and network dependence

On the primary HUMSAVAR set, mean AUC across three seeds was 0.6932 for raw ESM, 0.7304 for STRING FPP and 0.7025 for BioGRID FPP. STRING-minus-raw ensemble differences were +0.0375 AUC (95% CI [+0.0173, +0.0564]) and +0.0463 AUPRC ([+0.0151, +0.0745]). The stricter nested subset supported the same direction: +0.0352 AUC ([+0.0034, +0.0667]) and +0.0550 AUPRC ([+0.0111, +0.0962]). BioGRID-minus-raw intervals crossed zero on both sets (Fig. 3A; Supplementary Tables 16–17).

On the original ClinVar split, BioGRID representation-only mean AUC was 0.7592, compared with 0.7835 for raw ESM and 0.7950 for STRING. BioGRID fusion improved over scores alone (ensemble ΔAUC +0.0124, CI [+0.0034, +0.0210]), but its incremental margin over raw-embedding fusion remained uncertain (+0.0009, [-0.0011, +0.0036]). BioGRID fusion was below STRING fusion in both original and homology splits under the fitted-ensemble intervals (Fig. 3B). In the homology split, its AUC interval against scores alone ended just below zero, while the AUPRC interval crossed zero. These exploratory comparisons support network dependence, not a universal advantage or consistent harm from physical-network propagation.

D. Exploratory topology and prediction changes

Historical retained-reference predictions showed non-monotonic gains across topology strata (Supplementary Fig. S2 and Table 5). All 53 joint cells and highlighted profiles were inspected on the same test set; their intervals do not account for subgroup selection. Historical corrected/degraded predictions also depended on the decision threshold and uncalibrated scores (Supplementary Fig. S1 and Tables 6 and 10). These descriptions generate hypotheses rather than validated subgroup or clinical decision rules.

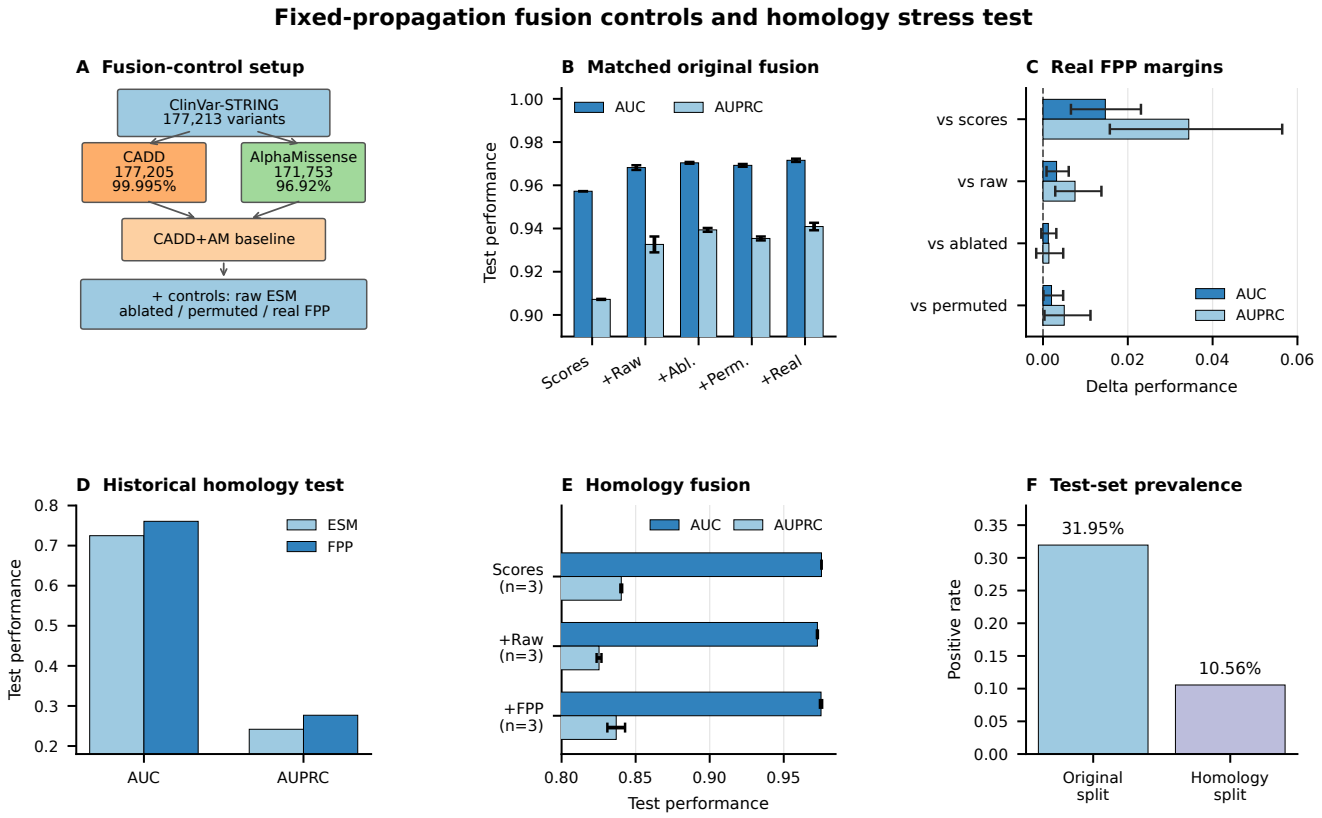


Fig. 2. Matched fusion and homology-aware evaluation. A, External-score coverage and fusion setup. B, Original-split matched fusion, mean and SD over three seeds per arm; permutation uses three graphs by three seeds. C, Real-FPP minus control performance on ensemble-averaged probabilities, with paired protein-clustered 95% bootstrap intervals (2,000 replicates). Ensemble point differences can differ from differences of mean run metrics in B. Intervals condition on fitted models and sampled graph permutations; they omit retraining and new-permutation uncertainty. D, Historical retained homology-aware representation estimates. E, New homology-aware fusion: all three arms use the same three seeds, classifier protocol and maximum 50 epochs; error bars indicate SD. F, Test prevalence by split. Scores denotes CADD+AlphaMissense; Raw, Abl., Perm. and Real denote added representations.

IV. Discussion

Matched comparisons support an evaluation-dependent effect of fixed propagation. On the original ClinVar representation comparison, mean performance improved over raw embeddings, but protein-level intervals for that gain crossed zero. Graph interventions show sensitivity to connectivity and protein–topology correspondence; edge removal also changes normalization and leaves indirect paths, so it does not isolate a causal connectivity effect. The transductive design permits held-out protein features and graph structure, without held-out labels.

The homology-aware STRING fusion results do not support a claim of consistent graph-induced deterioration. Three matched seeds resolve the previous unequal repeat count, while uncertainty remains around the STRING-versus-score comparison. Redundant high-dimensional features and changes in protein composition offer plausible explanations, rather than established mechanisms. The frozen partition and differing prevalence limit broader cross-split conclusions.

External HUMSAVAR evaluation supports transfer of STRING protein-context features under explicit variant and protein exclusions. BioGRID strengthens the audit by showing that this benefit does not automatically extend to another graph. However, STRING incorporates evidence from multiple sources and overlaps BioGRID; their density, coverage, release and evidence composition differ. This comparison cannot isolate a causal effect of network identity or show that physical interactions are unhelpful.

HUMSAVAR and ClinVar may share underlying literature or annotations, so an external curated source is not an independent patient cohort or prospective validation. The strict set is nested, not a second replication; full homology separation and external-predictor development overlap were not excluded. Protein-only outputs cannot distinguish substitutions within a protein, and HUMSAVAR substitutions need not all be genomic SNVs. External clinical-score fusion was not evaluated. Fixed propagation does not establish what learned graph models could achieve. Selected topology profiles require independent confirmation; clinical use would additionally require

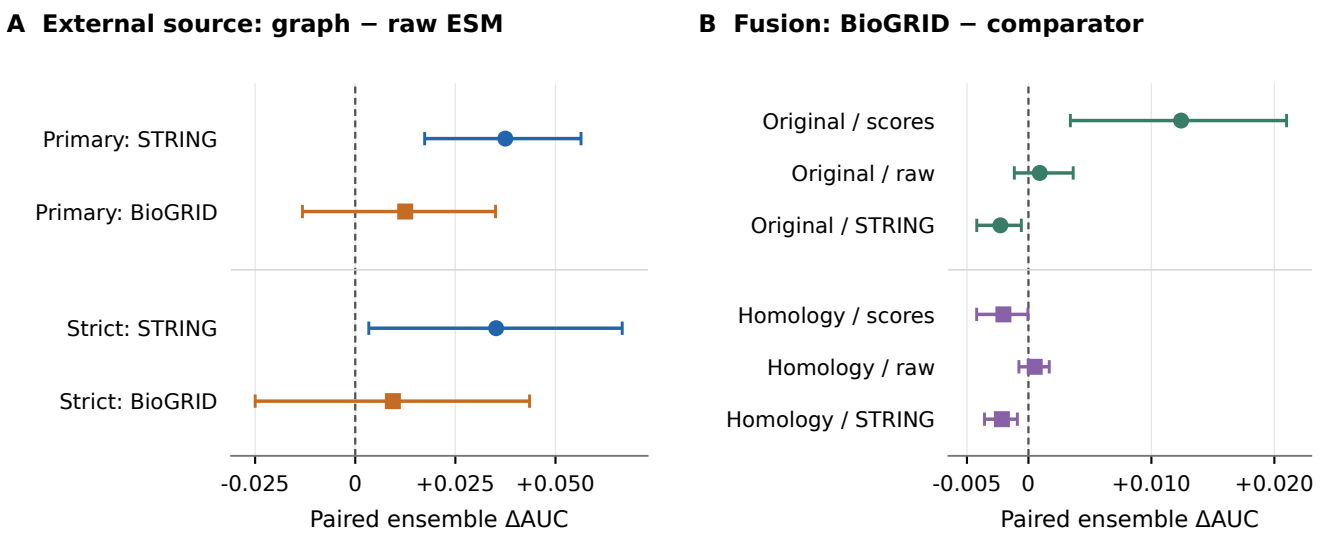


Fig. 3. External-source validation and alternative-network sensitivity. A, STRING- or BioGRID-FPP minus raw-ESM AUC on the primary HUMSAVAR set (10,785 variants; 2,498 proteins) and its stricter nested subset (6,934; 1,713). B, BioGRID fusion minus scores alone, raw-ESM fusion or STRING fusion, within each original or homology split. Points are differences in AUC of three-seed mean probabilities; bars are paired protein-clustered 95% bootstrap intervals (2,000 resamples), conditional on fitted models. The dashed line marks zero. Axes use different scales. Strict is a sensitivity subset, not an independent replication. Intervals are exploratory and not adjusted for multiple comparisons.

calibration and decision-rule validation.

a) Practical recommendations: Compare graph features with both raw protein embeddings and strong score baselines. Use the same training protocol across controls and report whether held-out proteins are visible in the graph. Quantify uncertainty at the protein level and check gains within the intended split. Treat selected subgroups as hypotheses, and evaluate calibration before using prediction changes to guide decisions.

V. Conclusion

Fixed STRING propagation shows evaluation-dependent gains and supported external-source transfer under the specified exclusions. BioGRID comparisons demonstrate sensitivity to network choice, with uncertain incremental benefit over raw-embedding fusion. Matched controls, strong baselines and protein-level uncertainty are essential; these findings do not establish performance limits for trained graph models or clinical decision validity.

Data and Code Availability

Code, dated manuscript revisions and both matched and external-validation records are available at https://github.com/brynn-t02/ppi-graph-missense-audit. The archive includes environment specifications, executed protocols, compact results and input checksums, with retained predictions for the earlier matched audit. Large provider resources, tensors and new checkpoints/prediction arrays are retained separately rather than redistributed. Reconstruction uses ClinVar, STRING, CADD, AlphaMissense, UniProt and BioGRID resources.

AI-Tool Disclosure

OpenAI ChatGPT, OpenAI Codex and Anthropic Claude supported literature organization, analysis-code drafting and language editing. Historical experiments were performed by the authors. Revision-stage scripts assisted by these tools restored an isolated runtime, checked input provenance, trained matched downstream classifiers, assembled the external-source and alternative-network validation, and recomputed sensitivity statistics. Actual logs, fitted predictions and checksums are retained for review. The authors retain responsibility for study design, interpretation, citation verification and final approval. The tools are not authors; no synthetic outcomes were substituted for experimental results.

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